Autonomous Vehicles

1. Perception Sensors

How does a lidar work, and what data does it give?

A LiDAR fires quick, sweeping pulses the bounce off objects within field of view and returns to it, The distance is calculated by the formula $\frac{Time \times Speed\ of\ Light}{2}$. The LiDAR eventually gives a 3D point cloud, each point is later used to identify cone color through intensity of the point on the point cloud.

How does a camera work, and what data does it give?

A Camera takes light and emits it onto a chip that uses that light to generate an array of 2D pixels with varying intensities of RGB. It gives a 2d pixels image that could be passed in our case through a CNN to define a color class, and keypoint regression to define the 3 properties of a cone through a 2D image.

How does a radar work, and what data does it give?

It sends out radio waves and listens for the bounce. The delay tells it how far away something is, and a slight change in the returning wave (Doppler shift) tells it how fast that thing is moving. It gives a short list of distances, speeds and rough directions with a drawback of being more sensitive to noise.

2. Stereo Vision

How can two cameras give the 3D position of an object?

Given a known gap between two cameras that view the same object at different angles, the image would shift more sensitively between the two images the closer the object gets, so using the gap, and the shifting between the images we can use a formula to determine the distance from the object, coupled with the current position of the perceiving vehicle, we can deduce the final 3D position on the shared map.

3. Machine Learning

What are neural networks?

They are a chain of very simple units, each taking numbers in, weighting them, applying a specific formula and passing the result on. Thousands of units stacked in layers can represent almost any pattern. The weights are what the network knows.

How do they learn?

They start with random weights, so the first guesses are meaningless. Each guess is compared with the right answer and scored on how wrong it was. The network then works out how much each weight contributed to the mistake and nudges every one slightly in the better direction using backpropagation which is the learning method of any ANN, repeating over millions of examples until the guesses stop improving.

How can they help process sensor data for understanding the surroundings?

They can take the cone images from the LiDAR sensor (3D point cloud) to estimate the color of the cone in question along with a confidence/probability index.

With Cameras they can be used with multiple object detection and keypoint regression to match a 2D image of a cone to its estimated 3D properties

4. Kalman Filter

What is a Kalman filter, and how is it used to work out where a vehicle is?

It is a method for tracking something you cannot measure cleanly. It combines a guess of what should happen next with what the sensors actually report, weighting each by how trustworthy it is, so the result is better than either alone. It repeats this process, and the corrected answer becomes the starting point for the next prediction.

What would you track, and which sensors would you use?

Track where the car is and which way it faces, how fast it is moving and turning, how quickly it is speeding up, how much each wheel is slipping, and the slow errors building up in the motion sensor. Use a motion sensor for turning and acceleration, a ground-speed sensor for true speed, wheel sensors for wheel rotation, and the cameras and laser for landmarks.

5. SLAM

What is SLAM?

Slam is the initial map forming operating mode in an unknown track that tracks vehicle motion while simultaneously building a shared map that could be used by other modes in later tracks

What are the different types?

They differ by what they sense (camera, laser, or both alongside a motion sensor), by how the map is stored, and by method. One method keeps only the current best guess and updates it constantly; the other stores the whole journey and later rearranges it so everything fits at once.

How is a map built, and which sensors help?

Spot landmarks and note where they sit relative to you, allow for the delay since they were seen, move your position guess forward using speed, then decide whether each landmark is new or already known and add or refine it. A laser gives accurate shapes, cameras give cheap detail and colour, and motion and wheel sensors fill the gaps between sightings.

How does SLAM work out where the vehicle is? (first type)

The first type keeps thousands of parallel guesses running at once, each with its own position and its own map. Every time something is seen, guesses whose maps fail to predict it are thrown away and the good ones are copied. A successful map converges in the end.

How does SLAM work out where the vehicle is? (second type)

The second type stores every past position along with every measurement linking them together. It then solves for the arrangement that best satisfies all of those measurements at the same time. So when it recognizes an old place, the correction is spread backwards over the whole journey instead of just being applied at the end.

6. Graph Search Algorithms

How does BFS work, and what are its pros and cons?

It checks in layers and will always find the path with the fewest steps, but it does not consider the weight/effort of the steps themselves.

How does DFS work, and what are its pros and cons?

It follows a certain path as deep as that path goes, covering all of its branches and backs off to another path when the first one is completely finished. This could be costly if the target is close but some paths are very long/incorrect. It might not always find the best path in either step count or weight.

How does Dijkstra's algorithm work, and what are its pros and cons?

It handles its selected route based on which next step requires the least total weight, which is more likely to produce an optimal path, but without a metric to measure the target against the next step, some moves may be redundant although the best path is likely to be found.

How does A work, and what are its pros and cons?*

It works like Dijkstra's with an addition of heuristics which are very helpful in deciding the proper next step. This, however, is more demanding memory wise than Dijkstra's, the heuristic must be carefully picked and might produce an invalid path in case there is an obstacle/barrier in that path. The target must also be known relative to each node to measure the heuristic value of each node against it which is not always available.

7. Practical Motion Planning

A cleaning robot's paths are too rough and zig-zag. What is the solution?

The path is jagged because the robot is only allowed to step in eight fixed directions, so anything diagonal comes out as a staircase. Straighten it afterwards by deleting any corner you can cut with a clear straight line.

What is the risk of that solution?

The straightened route no longer runs through the squares that were checked as safe, so it can clip a wall or a table leg, or pass much closer to things than intended. In tight spots the zig-zag may have been the only way through, and straightening it can cut right through a doorframe. Everything smoothed has to be re-checked against the map, with the original route kept as a fallback.

8. Control Theory and Philosophy

What is control, and what is control theory? Why does it matter?

Control is making something behave the way you want by deciding what the heater or motor should do moment by moment. The theory is that the control system operates through feedback to decide what happens next while remaining stable, responsive, and in line with the expected outcome despite abnormalities and edge cases. It matters the most for automated critical systems like lifts and aircrafts and driverless cars where one mistake could be fatal.

What is the difference between open-loop and closed-loop control?

Open loop follows a fixed plan and never checks the result, a simple, cheap and straightforward process that is not affected no matter what the current outcome could be. Closed loop measures the result through feedback loops, compares it with the target, and acts on the difference to stabilize, act on, or increase the intensity of the operation like an AC unit. Open loop is simpler and cheaper. closed loop corrects itself but needs sensors and can start swinging if mishandled, which can lead to worse results than an open loop.

What is feedback?

Feedback means sending information about the result back to the start, so the next decision is based on how far off you actually are. It is what makes a system correct itself, and its great strength is that it does not need to know why things went off target. it needs sensors, and delayed or noisy readings can make it useless.

Imagine a situation where you need feedback.

Holding a steady 100 km/h on the motorway. A fixed throttle position works on flat road, but the car slows on a climb, speeds up going downhill, and drops with a headwind or extra passengers. Measuring the actual speed and adjusting the throttle from the gap holds it steady in those cases.

9. PID Control

What is a PID controller, and what do its three parts do?

It watches the gap between where you are and where you want to be, and builds its output from three parts. The first does the main pushing, but always settles slightly short of the target. The second builds up over time and removes that leftover shortfall, at the cost of being slow and overshooting. The third watches how fast the gap is closing and eases off early to prevent overshoot, but it exaggerates any jitter in the readings.

What is the effect of each part, and how do you tune them?

More of the first is faster but shakier, more of the second removes the leftover error but overshoots, and more of the third steadies things until noise takes over. Tuning by hand: raise the first until it's just short of the output, then add the second to remove the shortfall and the third to calm the overshoot. The classic recipe instead pushes the system until it swings steadily, notes that setting and the swing time, and reads the three values off a standard table.

How would you design a temperature control system for an industrial furnace?

A temperature probe in a protective sleeve reports the heat, the controller compares it with the target, and sets the power unit feeding the heating elements anywhere from off to full. Because a furnace heats quickly but cools only slowly, overshoot has to be avoided: raise the target gradually rather than in one jump, and stop the middle part building up while the power is stuck at full, or it will badly overshoot on the way up. Add an independent cut-out in case the probe fails.

What are the limitations of PID, and when is something else better?

It handles one measurement with one output, so it cannot manage several things that affect each other. It knows nothing about limits it must not cross, and it only reacts to what has already happened rather than what is coming. When limits, interacting parts, or known upcoming changes matter, a method that plans a few seconds ahead fits better — which is what the racecar uses once it knows the track.